

Theorem 1. Let e_I and f_I be supported on $(2n+1)I$. Let $r > 2$ and suppose that

$$\sup_I w(I)^{\frac{1}{2}-\frac{1}{r}} \|e_I\|_{L_r(\mathbb{R}^N, w(x)dx)} \leq 1, \cdot \sup_I w(I)^{\frac{1}{2}-\frac{1}{r}} \|f_I\|_{L_r(\mathbb{R}^N, w(x)dx)} \leq 1.$$

If V is an integral operator with integral kernel

$$K : (x, y) \rightarrow \sum_I \lambda_I e_I(x) f_I(y),$$

then

$$\|V\|_{p,\infty} \leq c_{p,n,r} \|\lambda\|_{p,\infty}.$$

Theorem 2. Integral kernel $K : (x, y) \rightarrow K(x, y)$, $x, y \in \mathbb{R}^N$, of $\Delta^{-\frac{1}{2}}$, where Δ is Dunkl Laplacian is smooth except on the sets $\{x = gy\}$, $g \in G$, or on the planes $\{\langle x, v \rangle = 0\}$, $\{\langle y, v \rangle = 0\}$.

Proof. **IT IS EXACTLY THIS WE CANNOT PROVE YET.** □

Lemma 1. We have

$$K(\lambda x, \lambda y) = \lambda^{-M} K(x, y), \quad x, y \in \mathbb{R}^N.$$

Here,

$$M = N - 1 + \sum_{v \in R} k(v).$$

Proof. Indeed, let D_λ be the dilation by λ . It is obvious (just like in the Euclidean case) that

$$D_\lambda^{-1} T_\xi D_\lambda = \lambda^{-1} T_\xi.$$

Thus,

$$D_\lambda^{-1} \Delta D_\lambda = \lambda^{-2} \Delta.$$

Thus,

$$\Delta^{-\frac{1}{2}} D_\lambda = \lambda D_\lambda \Delta^{-\frac{1}{2}}.$$

In other words,

$$\int_{\mathbb{R}^N} K(x, y) \xi(\lambda^{-1} y) w(y) dy = \lambda \int_{\mathbb{R}^N} K(\lambda^{-1} x, y) \xi(y) w(y) dy.$$

Substituting $y = \lambda z$ in the first integral and $y = z$ in the second one, we write

$$\lambda^{N+\sum_{v \in R} k(v)} \int_{\mathbb{R}^N} K(x, \lambda z) \xi(z) w(z) dz = \lambda \int_{\mathbb{R}^N} K(\lambda^{-1} x, z) \xi(z) w(z) dz.$$

In other words,

$$\lambda^{N+\sum_{v \in R} k(v)} K(x, \lambda z) = \lambda K(\lambda^{-1} x, z).$$

Substituting λx instead of λ and y instead of z , we conclude the argument. □

Lemma 2. We have (provided that R spans $\mathbb{R}^N$)

$$f(x) K(x, y) = c \sum_{k,l \in \mathbb{Z}^N} c_{k,l} \sum_{I \in \mathcal{D}} s(I)^{-M} w(I)^{1-\frac{1}{r}} \|f\|_{L_r(I)} e_{k,I}(x) f_{l,I}(y),$$

where $e_{k,I}$ and $f_{l,I}$ are supported in $(2n+1)I$ *can we specify n ?*

$$\sup_{k,I} w(I)^{\frac{1}{2}-\frac{1}{r}} \|e_{k,I}\|_{L_r(\mathbb{R}^N, w(x)dx)} \leq 1,$$

$$\sup_{l,I} w(I)^{\frac{1}{2}-\frac{1}{r}} \|f_{l,I}\|_{L_r(\mathbb{R}^N, w(x)dx)} \leq 1.$$

Proof. Apply Whitney decomposition to the set $(\Gamma \times \Gamma) \setminus \{x \neq y\}$ (we do not even need the linear transformations). Here, each cube $Q \in \mathcal{F}$ is of the shape $I \times J$, where $s(I) = s(J)$. Distance from $x \in I$ to the boundary of Γ is comparable to $s(Q) = s(I)$. Distance from $y \in J$ to the boundary of Γ is comparable to $s(Q) = s(I)$. Distance from $x \in I$ to $y \in J$ is comparable to $s(Q) = s(I)$. Hence, there exists a fixed number n (possibly, dependent on Γ) so that $J \subset (2n+1)I$.

We have (provided that R spans $\mathbb{R}^N$)

$$|\langle x, v \rangle| \geq \epsilon s(I), \quad |\langle y, v \rangle| \geq \epsilon s(I), \quad |x-y| \geq \epsilon s(I), \quad |x|, |y| \leq \epsilon^{-1} s(I), \quad x \in I, \quad y \in J.$$

Set

$$\Omega_\epsilon = \{(x, y) \in \Gamma : |\langle x, v \rangle| \geq \epsilon^2, \quad |\langle y, v \rangle| \geq \epsilon^2, \quad |x-y| \geq \epsilon^2, \quad |x|, |y| \leq 1\}.$$

Thus,

$$\left(\frac{x}{\epsilon^{-1}s(I)}, \frac{y}{\epsilon^{-1}s(I)}\right) \in \Omega_\epsilon, \quad x \in I, \quad y \in J.$$

By Theorem 2, $K \in C^m(\Omega_\epsilon)$ for every $m \in \mathbb{N}$. Hence, there exists a rapidly decreasing sequence $\{c_{k,l}\}_{k,l \in \mathbb{Z}^N}$ such that

$$K(x, y) = \sum_{k,l \in \mathbb{Z}^N} c_{k,l} e^{i\langle k, x \rangle} e^{i\langle l, y \rangle}, \quad (x, y) \in \epsilon^{-1}\Omega_\epsilon.$$

We, therefore, have

$$\begin{aligned} K(x, y) &= (\epsilon^{-1}s(I))^{-M} K\left(\frac{x}{\epsilon^{-1}s(I)}, \frac{y}{\epsilon^{-1}s(I)}\right) = \\ &= (\epsilon^{-1}s(I))^{-M} \sum_{k,l \in \mathbb{Z}^N} c_{k,l} e^{\frac{i\epsilon\langle k, x \rangle}{s(I)}} e^{\frac{i\epsilon\langle l, y \rangle}{s(I)}}, \quad x \in I, \quad y \in J. \end{aligned}$$

Set $J_I = \cup_{I \times J \in \mathcal{F}} J$. Integral kernel of $M_f \Delta^{-\frac{1}{2}}$ is

$$\sum_{k,l \in \mathbb{Z}^N} c_{k,l} \sum_{I \in \mathcal{D}} (\epsilon^{-1}s(I))^{-M} f(x) e^{\frac{i\epsilon\langle k, x \rangle}{s(I)}} \chi_I(x) \cdot e^{\frac{i\epsilon\langle l, y \rangle}{s(I)}} \chi_{J_I}(y).$$

Set

$$e_{k,I}(x) = \frac{w(I)^{\frac{1}{r}-\frac{1}{2}}}{\|f\|_{L_r(I, w(x)dx)}} f(x) e^{\frac{i\epsilon\langle k, x \rangle}{s(I)}} \chi_I(x).$$

Obviously,

$$\sup_{k,I} w(I)^{\frac{1}{2}-\frac{1}{r}} \|e_{k,I}\|_{L_r(\mathbb{R}^N, w(x)dx)} = 1.$$

Set

$$f_{l,I}(x) = w(I)^{-\frac{1}{2}} e^{\frac{i\epsilon\langle l, x \rangle}{s(I)}} \chi_{J_I}(x).$$

Obviously,

$$\begin{aligned} \sup_{l,I} w(I)^{\frac{1}{2}-\frac{1}{r}} \|f_{l,I}\|_{L_r(\mathbb{R}^N, w(x)dx)} &= \sup_I w(I)^{-\frac{1}{r}} \|\chi_{A_I}\|_{L_r(\mathbb{R}^N, w(x)dx)} \leq \\ &\leq \sup_I w(I)^{-\frac{1}{r}} \|\chi_{(2n+1)I}\|_{L_r(\mathbb{R}^N, w(x)dx)} = \left(\sup_I \frac{w((2n+1)I)}{w(I)}\right)^{\frac{1}{r}} = c_n < \infty. \end{aligned}$$

Integral kernel of $M_f \Delta^{-\frac{1}{2}}$ is

$$\sum_{k,l \in \mathbb{Z}^N} c_{k,l} \sum_{I \in \mathcal{D}} (\epsilon^{-1}s(I))^{-M} w(I)^{1-\frac{1}{r}} \|f\|_{L_r(I, w(x)dx)} e_{k,I}(x) f_{l,I}(y).$$

□

By Theorem 1 and by the triangle inequality, the norm $\|M_f \Delta^{-\frac{1}{2}}\|_{N,\infty}$ is controlled by

$$\sum_{k,l \in \mathbb{Z}^N} |c_{k,l}| \cdot \left\| \left\{ s(I)^{-M} w(I)^{1-\frac{1}{r}} \|f\|_{L_r(I, w(x)dx)} \right\}_{I \in \mathcal{D}} \right\|_{N,\infty}.$$

Note that

$$w(x) \sim s(I)^{\sum_{v \in R} k(v)}, \quad x \in I.$$

Thus,

$$w(I) = \int_I w(x)dx \sim s(I)^{N + \sum_{v \in R} k(v)},$$

$$\|f\|_{L_r(I, w(x)dx)} \sim s(I)^{\frac{1}{r} \sum_{v \in R} k(v)} \|f\|_{L_r(I, dx)}.$$

So, $\|M_f \Delta^{-\frac{1}{2}}\|_{N,\infty}$ is controlled by

$$\begin{aligned} & \left\| \left\{ s(I)^{-M} s(I)^{(1-\frac{1}{r})(N + \sum_{v \in R} k(v))} s(I)^{\frac{1}{r} \sum_{v \in R} k(v)} \|f\|_{L_r(I, dx)} \right\}_{I \in \mathcal{D}} \right\|_{N,\infty} = \\ & = \left\| \left\{ s(I)^{-M} s(I)^{N + \sum_{v \in R} k(v)} s(I)^{-\frac{N}{r}} \|f\|_{L_r(I, dx)} \right\}_{I \in \mathcal{D}} \right\|_{N,\infty}. \end{aligned}$$

Since

$$M = N - 1 + \sum_{v \in R} k(v),$$

it follows that the latter expression is

$$\left\| \left\{ s(I)^{1-\frac{N}{r}} \|f\|_{L_r(I, dx)} \right\}_{I \in \mathcal{D}} \right\|_{N,\infty}.$$

By Zhijie lemma, this is controlled by $\|f\|_{L_d(\mathbb{R}^N)}$.